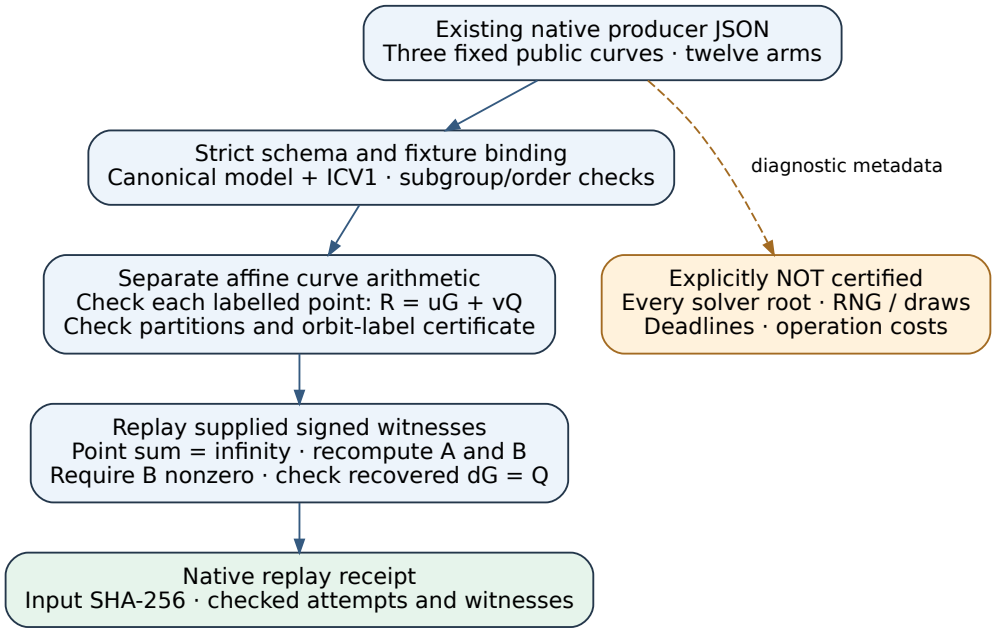
Independent APS transcript replay

Classical bounded correctness · 2026-10-29

Observed result. Fourteen actual native producer transcripts replay successfully across twelve supported arms and three fixed public curves. The same nine replay/corruption tests pass in both the standalone package and repository-example build. This is verification support, not a speedup or a new relation-finding method.

Verification contract

The replayer uses separate ordinary-residue affine arithmetic. It links serde, serde_json, clap and sha2, with no producer curve, solver, recovery or Montgomery implementation. It binds the fixture/model/ICV1 identity, counts the tiny curve's points, checks each labelled point, and replays the supplied coordinate/sign witnesses. Recovery requires the recomputed signed coefficients, nonzero B, the scalar congruence, and dG = Q. Orbit pruning additionally requires independently checked label propagation.



Boundary. A checked witness does not prove that the solver supplied every root. Sampling, draw history, deadlines, symbolic work and operation costs remain uncertified. A no-witness receipt reports zero recoveries; producer status and completeness flags do not become an absence proof.

Sources and provenance

APS revised manuscript, sections 3–3.1: known labels, signed zero sum, partition constraints and scalar recovery. Milne, Elliptic Curves, I section 3, pp.25–28: group law, inverses and infinity. Native interfaces and limitations are recorded in the earlier APS follow-up and complete-cost problem.

Tested producer base: `117c9c4f3ee85d4dc32f275893f6e50b591e4a47`. Per-input SHA-256 and native replay receipts are in validation/replays/; source hashes and actual check outcomes are in validation/summary.json. Third-party papers and broad build/infrastructure logs are not redistributed.

Corpus, controls and remaining work

Case	Fixture	Attempts	Sets	Witnesses
block-symmetric4	toy101	1	2	1
cover3	toy101	3	1	1
expanded4	toy101	1	1	1
hybrid3	toy101	4	1	1
hybrid4	toy101	1	1	1
line3-toy211	toy211	2	1	1
line3	toy101	3	1	1
orbit-line3	toy103_j0	1	9	1
orbit-pruned-line3	toy103_j0	1	1	1
orbit3	toy103_j0	1	3	1
partitioned3	toy101	8	2	1
split4-toy211	toy211	1	1	1
split4	toy101	2	2	1
unordered3	toy101	3	1	1

Counts are per recorded input. These are curated fixed correctness controls, not an unbiased yield population or a timing panel. Failed search attempts inside the emitted transcripts are retained and checked.

Exact curve identities

`ICV1:fp-101:-5:107:36:unk:unk:r:9ced75c2406a`
`ICV1:fp-211:-15:227:153:unk:unk:r:9004a75cb03c`
`ICV1:fp-103:7:97:0:unk:unk:r:394a8c88b017`

Negative controls and retained failures

Tests reject changed points, labels, scalar representatives, curve/model identities, targets, recovery coefficients and signs, coordinate membership, duplicate keys/sets and contradictory outcome flags. A special orbit control changes valid labels while preserving the points and selected witness; the independent propagation check refuses it.

Initial generation completed nine cases, then the default guaranteed-label orbit case hit the producer’s short/overlapping-orbit guard before JSON emission. Four later cases were not attempted. The three final orbit cases use the existing producer unit test’s random-label control with the same seed. The first nine files were preserved; five remaining cases completed. The refusal and initial lint correction remain in validation/failures.json. Pre-transcript refusals still require an outer experiment export.

Scope after this change

The bounded independent-witness replay gap is addressed. Full accounting, ecbench/export integration, broader parameter support, a frozen protocol, matched references, checkpoint authorization and controlled performance validation remain open. No performance run or historical Python research path was executed, and no full library-suite pass is claimed.

All nineteen historical archive hashes still match. Canonical IC leaderboard, scoreboard, progress timeline and performance-gains values are unchanged: this work supplies no admitted measurement or new curve route.